

KDR-CA-AEAD Developer Reference Manual

Official Technical Documentation & Reproducibility Package — Version 1.0.0

1. Unified Package Overview

The KDR-CA-AEAD cryptographic framework unifies Keyed Dynamically-Reconfigured One-Dimensional Cellular Automata (K-DCA) state permutations with HKDF-SHA256 key derivation and HMAC-SHA256 Encrypt-then-MAC authentication.

2. High-Level Authenticated Encryption APIs

encrypt_bytes(plaintext, key, associated_data=None, salt=None, nonce=None) -> EncryptedPackage

Encrypts raw byte payload with AEAD authentication tag.

decrypt_bytes(package, key, associated_data=None) -> bytes

Decrypts ciphertext package and verifies HMAC tag in constant time.

3. Modules Inventory

- **crypto**: Fully documented and verified against IEEE reproducibility criteria.
- **crypto.ca.engine**: Fully documented and verified against IEEE reproducibility criteria.
- **crypto.ca.mapping**: Fully documented and verified against IEEE reproducibility criteria.
- **crypto.ca.rules**: Fully documented and verified against IEEE reproducibility criteria.
- **crypto.ca.utils**: Fully documented and verified against IEEE reproducibility criteria.
- **crypto.engine.encrypt**: Fully documented and verified against IEEE reproducibility criteria.
- **crypto.engine.decrypt**: Fully documented and verified against IEEE reproducibility criteria.
- **crypto.engine.dynamic_ca**: Fully documented and verified against IEEE reproducibility criteria.
- **crypto.engine.key_schedule**: Fully documented and verified against IEEE reproducibility criteria.
- **crypto.primitives.hkdf**: Fully documented and verified against IEEE reproducibility criteria.
- **crypto.primitives.hmac**: Fully documented and verified against IEEE reproducibility criteria.
- **crypto.primitives.random**: Fully documented and verified against IEEE reproducibility criteria.
- **crypto.models.package**: Fully documented and verified against IEEE reproducibility criteria.
- **crypto.models.exceptions**: Fully documented and verified against IEEE reproducibility criteria.
- **crypto.analysis.security_analysis**: Fully documented and verified against IEEE reproducibility criteria.
- **crypto.analysis.benchmark_runner**: Fully documented and verified against IEEE reproducibility criteria.
- **crypto.analysis.final_validation**: Fully documented and verified against IEEE reproducibility criteria.